Progress Report #1: The Effect of Immigration on House Prices during the Age of Mass Migration

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Abstract

This report documents the first round of empirical analysis for a project studying the effect of European immigration on US house prices during the Age of Mass Migration (1910–1930). We combine Tabellini (2020) city-level panel data and instrumental variable strategy with historical house price data from Lyons et al. (2024). Using predicted immigrant settlement patterns based on pre-1900 distributions as an instrument, we estimate 2SLS regressions on a sample of 26 US metro areas observed across three decades. The first stage is strong (Kleibergen-Paap F = 90.6), but the second stage yields a null result: the 2SLS coefficient on immigration is −0.70 with a standard error of 1.84 (p = 0.70). We discuss the limitations of the small sample and outline next steps.

*University of Cologne. This project uses Claude Code (Anthropic) as an AI research assistant. The research question, identification strategy, and data source decisions were made by the human researcher. Claude Code implemented the data pipeline, wrote Stata scripts, ran regressions, and drafted this report. See Section 7 for a detailed attribution.

1 Research Question

Does immigration affect house prices? This is a question with direct policy relevance today, but identification is difficult because immigrants choose where to live based on local economic conditions that also affect housing markets. We exploit a well-known historical setting — the Age of Mass Migration to the United States (1900–1930) — where a credible instrumental variable strategy exists.

Specifically, we ask: **did European immigration to US cities between 1910 and 1930 causally affect local house prices?**

We build on Tabellini (2020), who studies the effects of immigration on public spending, taxation, labor markets, and political outcomes using a shift-share instrument based on pre-1900 immigrant settlement patterns. We replace his outcome variables with historical house prices from Lyons et al. (2024), who construct house price series for 30 US cities from 1890 to 2006 using newspaper advertisements.

2 Data

2.1 Tabellini City-Level Panel

We use the city-level dataset from Tabellini (2020), which contains 180 US cities observed at decennial frequency (1910, 1920, 1930). The key variables are:

- **Endogenous variable:** Predicted fraction of recent European immigrants relative to predicted city population (`recentEU_predpop`).
- **Instrument:** Predicted immigrant settlement based on the pre-1900 geographic distribution of European immigrants across MSAs (`ZEUMSA_predpop`). This is a leave-one-out version of the standard shift-share (Bartik) instrument.
- **Controls:** Log city population in 1900 interacted with year dummies, log immigration in 1900 interacted with year dummies, and log manufacturing value added in 1904 interacted with year dummies.

2.2 Lyons Historical House Prices

Lyons et al. (2024) construct house price and rent series for 30 major US cities from 1890 to 2006 using a database of 2.7 million newspaper real estate advertisements. We load the raw transaction-level data and collapse it to city-year level mean and median log sale prices

and log rents. We deflate nominal prices using the Officer-Williamson CPI series (base year = 1910). To match the decennial structure of the Tabellini data, we average prices within three-year windows centered on census years (e.g., 1909–1911 for the 1910 decade).

2.3 Merged Sample

We merge the two datasets on IPUMS metropolitan area codes (`metarea`). Of the 30 Lyons cities and 180 Tabellini cities, **26 metro areas match across both datasets**, yielding a balanced panel of **78 observations** (26 cities × 3 decades). The matched cities are major urban centers including New York, Chicago, Philadelphia, Detroit, Boston, Los Angeles, San Francisco, and others.

Table 1: Summary Statistics

	N	Mean	SD	Min	Max
Panel A: Immigration Variables					
Fraction EU immigrants	78	0.151	0.101	0.011	0.398
Fraction recent EU immigrants	78	0.039	0.040	0.001	0.184
Predicted recent EU (endogenous)	78	0.044	0.047	0.001	0.202
Predicted EU settlement (instrument)	78	0.033	0.032	0.002	0.149
City population	78	752,796	1,187,149	37,700	6,930,400
Panel B: House Prices					
Real log sale price (1910 base)	78	8.103	0.357	7.334	9.099
Nominal log sale price	78	8.520	0.404	7.766	9.672
Number of sale transactions	78	1,040	756	66	3,037
Real log rent (1910 base)	78	3.175	0.289	2.577	4.152
Panel C: Tabellini Outcomes (for comparison)					
Per capita expenditures	78	17.560	8.133	7.290	43.642
Property tax rate	78	30.341	17.098	10.390	106.250
Employment rate	78	0.861	0.067	0.679	0.940
Log occupational score	78	3.243	0.040	3.116	3.316

Notes: Sample consists of 26 US metro areas observed in 1910, 1920, and 1930. Immigration variables are from Tabellini (2020). House prices are mean log prices from newspaper advertisements (Lyons et al., 2024), averaged within 3-year windows around census years and deflated by CPI (1910 = 1). Number of sale transactions is the total within each 3-year window.

3 Empirical Strategy

We follow Tabellini (2020) exactly, replacing his outcome variables with house prices. The estimating equation is:

$$y_{ct} = \beta \cdot \text{ImmShare}_{ct} + \alpha_c + \delta_t + \varepsilon_{ct} \tag{1}$$

where y_{ct} is the real mean log sale price in city c and decade t, ImmShare_{ct} is the predicted fraction of recent European immigrants, α_c are city fixed effects, and δ_t are year fixed effects. Standard errors are clustered at the metropolitan area level.

The endogeneity concern is that immigrants select into cities with favorable economic conditions, which may independently affect house prices. We instrument ImmShare_{ct} with the shift-share instrument Z_{ct}, which predicts current immigration using the pre-1900 geographic distribution of immigrants from each European origin country interacted with aggregate immigration flows from those countries in each decade.

The **first stage** is:

$$\text{ImmShare}_{ct} = \pi \cdot Z_{ct} + \alpha_c + \delta_t + \eta_{ct} \tag{2}$$

Given the small sample (78 observations, 26 city FE, 3 year FE), we use year fixed effects as the baseline specification. State-year fixed effects (Tabellini's preferred specification) are included as a robustness check, but they consume most degrees of freedom.

4 Results

4.1 First Stage

Table 2 reports the first stage. The instrument strongly predicts immigration in our subsample: the coefficient is 1.10 (SE = 0.12, $p < 0.001$) with a Kleibergen-Paap F-statistic of 90.6, well above the Stock and Yogo (2005) threshold of 16.38 for 10% maximal IV size. The first stage remains strong with state-year fixed effects (Column 2), though it weakens when adding baseline controls (Columns 3–4, F $\approx$ 4.9), suggesting collinearity between the controls and the instrument in this small sample.

4.2 Main Results

Table 3 reports the main results. The key finding is in Column 4: the 2SLS estimate of the effect of immigration on real log house prices is **−0.70 with a standard error of 1.84** (p = 0.70). The effect is economically small and statistically indistinguishable from zero.

Table 2: First Stage: Instrument on Predicted Immigration

	(1) Year FE	(2) State-Year FE	(3) St-Yr + Controls	(4) All Controls
Predicted EU settlement (Z)	1.102***	1.306***	1.559***	1.527***
	(0.117)	(0.079)	(0.238)	(0.262)
Observations	78	78	78	75
Within R^2	0.748	0.992	0.994	0.997
KP F-statistic	90.6	278.1	4.9	4.5
City FE	Yes	Yes	Yes	Yes
Baseline controls	No	No	Yes	Yes
Value added controls	No	No	No	Yes

Notes: Dependent variable is the predicted fraction of recent European immigrants relative to predicted city population. The instrument is the leave-one-out predicted settlement pattern based on pre-1900 geographic distributions. Standard errors clustered at the metro area level in parentheses. Baseline controls: log city population in 1900 and log immigration in 1900, each interacted with year dummies. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The OLS estimates tell a different story. Without controls (Column 1), OLS yields a small positive coefficient of 0.47 (insignificant). With baseline controls (Column 2), the OLS estimate becomes 2.83 (SE = 0.95, p = 0.006). The discrepancy between OLS-with-controls and 2SLS suggests that the positive OLS association is driven by immigrants selecting into economically booming cities — precisely the endogeneity that the IV addresses.

With state-year fixed effects (Columns 7–8), the 2SLS coefficient becomes negative and significant (-2.73, SE = 1.32, p = 0.04; and -8.20 with controls). However, the state-year FE specification is unreliable in this small sample: 19 of 60 state-year dummies are dropped due to collinearity, and the results with controls (Column 8) show signs of overfitting (coefficient magnitude triples).

Robustness checks using nominal prices, median prices, log rents, and dropping New Orleans (an outlier in Tabellini's analysis) all yield insignificant 2SLS coefficients in the range -0.72 to $+0.87$.

5 Limitations

1. **Small sample size.** The core limitation is having only 78 observations. With 26 city fixed effects and 2 year dummies, only 49 residual degrees of freedom remain for estimation. This severely limits statistical power: the 95% confidence interval for the baseline 2SLS estimate spans roughly -4.3 to $+2.9$ — we cannot rule out large positive

Table 3: Effect of Immigration on House Prices

	(1) OLS	(2) OLS	(3) RF	(4) 2SLS	(5) 2SLS	(6) 2SLS
Panel A: Year Fixed Effects						
Recent EU immigration	0.466	2.831***		−0.702	3.384**	2.999*
	(1.090)	(0.951)		(1.837)	(1.495)	(1.573)
Predicted EU settlement (Z)			−0.774			
			(2.136)			
KP F-statistic				90.6	4.9	4.5
Baseline controls	No	Yes	No	No	Yes	Yes
Value added controls	No	No	No	No	No	Yes
Panel B: State-Year Fixed Effects						
	(7) 2SLS	(8) 2SLS				
Recent EU immigration	−2.728**	−8.204***				
	(1.324)	(2.641)				
KP F-statistic	278.1	44.2				
Baseline controls	No	Yes				
Observations	78	78	78	78	78	75
City FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Dependent variable is real mean log sale price (1910 CPI base). Immigration is instrumented with predicted EU settlement patterns based on pre-1900 distributions. Panel A uses year fixed effects (baseline); Panel B uses state-year fixed effects. Baseline controls: log city population and log immigration in 1900, each interacted with year dummies. Value added controls: log manufacturing value added in 1904 interacted with year dummies. Standard errors clustered at the metro area level in parentheses. In Panel B, 19 of 60 state-year dummies are dropped due to collinearity. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

or negative effects.

2. **House price measure.** We use simple mean log sale prices from newspaper advertisements rather than a hedonic price index that controls for housing characteristics. Composition changes in the types of houses advertised could bias the city-year price averages.

3. **Decennial aggregation.** Collapsing annual house price data to decennial observations discards within-decade variation that could improve precision.

4. **Weak instrument with controls.** The first stage F-statistic drops below conventional thresholds when adding baseline controls, raising weak instrument concerns for the specifications with controls.

5. **Geographic coverage.** The 26 matched cities are disproportionately large metropolitan areas. Results may not generalize to smaller cities that received immigrants.

6 Next Steps

1. **Exploit annual variation.** Use annual Lyons price data (rather than decennial averages) to increase the effective sample size. This requires adapting the Tabellini specification to a city-year panel, possibly using interpolated or time-varying instruments.

2. **Hedonic price index.** Run the full Lyons HPI pipeline to construct quality-adjusted house price indices as the outcome variable.

3. **Heterogeneity.** Examine whether effects differ between high- and low-immigration cities, or between sale prices and rents (which may respond differently to demand shocks).

4. **Validate the subsample.** Replicate Tabellini's main results (spending, taxes, employment) in our 26-city subsample to verify that the null house price result is not driven by sample selection.

5. **Additional outcomes.** Explore construction activity, housing permits, or homeownership rates as complementary outcomes that may be more responsive to immigration-driven demand shocks.

7 Human–AI Collaboration

This project is conducted as a collaboration between a human researcher and Claude Code (Anthropic), an AI coding assistant. Transparency about the division of labor is a deliberate goal of the project.

Human researcher (Guohui Jiang):

- Defined the research question: effect of immigration on house prices during the Age of Mass Migration
- Selected the data sources: Tabellini (2020) replication package and Lyons et al. (2024) house price data
- Chose the identification strategy: replicate Tabellini's IV approach with house prices as outcome
- Specified the empirical design: city-decade panel, same controls and fixed effects, clustered standard errors
- Directed the overall workflow: build before analysis, variable choices, robustness checks
- Reviewed and interpreted all results

Claude Code (AI assistant):

- Explored both datasets to identify variable names, panel structure, and geographic identifiers
- Designed and implemented the city crosswalk (IPUMS metarea codes)
- Wrote all five Stata scripts: data exploration, Lyons data processing (2.7M observations), merge, summary statistics, and regressions
- Debugged execution issues: variable name mismatches, missing directories, Stata batch mode on Windows
- Ran the full pipeline and verified all outputs
- Drafted this progress report and the research log entries

Key observation: The AI assistant was effective at the mechanical parts of the workflow — data exploration, script writing, debugging, and execution. The human researcher provided the intellectual direction: what question to ask, what data to use, and what identification strategy to employ. The AI flagged practical issues (e.g., small sample size, degrees-of-freedom constraints) that informed the choice of year FE over state-year FE as the baseline specification.

References

Lyons, R., Shertzer, A., Gray, R., and Agorastos, N. (2024). "Historical House Prices in US Cities." *Quarterly Journal of Economics*.

Stock, J.H. and Yogo, M. (2005). "Testing for Weak Instruments in Linear IV Regression." In *Identification and Inference for Econometric Models*, Cambridge University Press.

Tabellini, M. (2020). "Gifts of the Immigrants, Woes of the Natives: Lessons from the Age of Mass Migration." *Review of Economic Studies*, 87(1), 454–486.